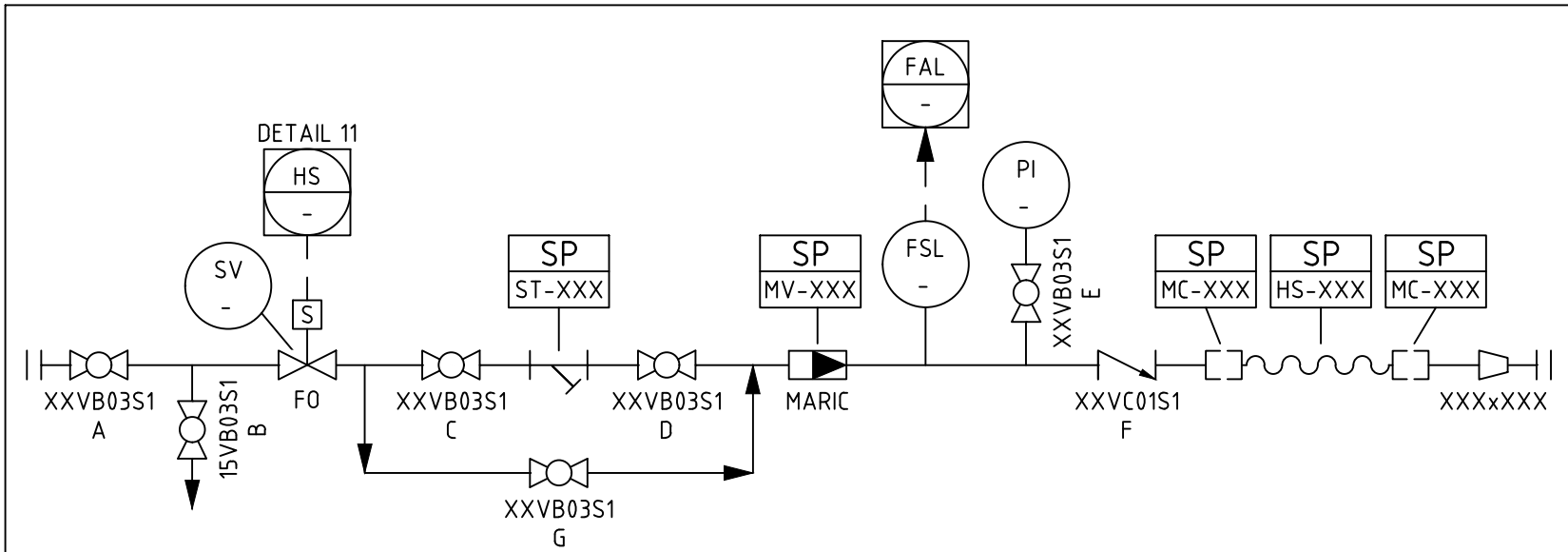


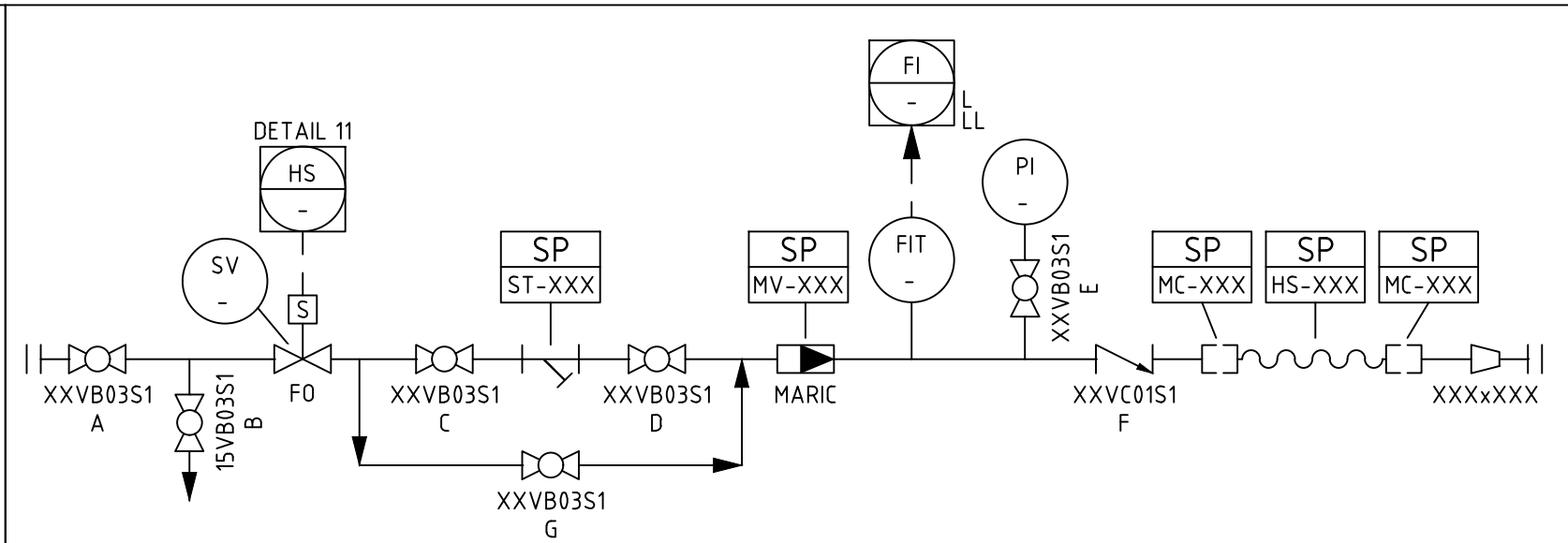
D:\CAD PROJECTS\235 NORTHERN STAR RESOURCES LTD\23517 KCGM GROWTH PROJECT\03 DELIVERABLES\PR\PID\ISSUED\12.09.25 - SIGNING\NATIVES\2000-F24-PID-PR-10035_1.DWG 09SEP25 8:22AM ANNE HOYNE

NOTES:

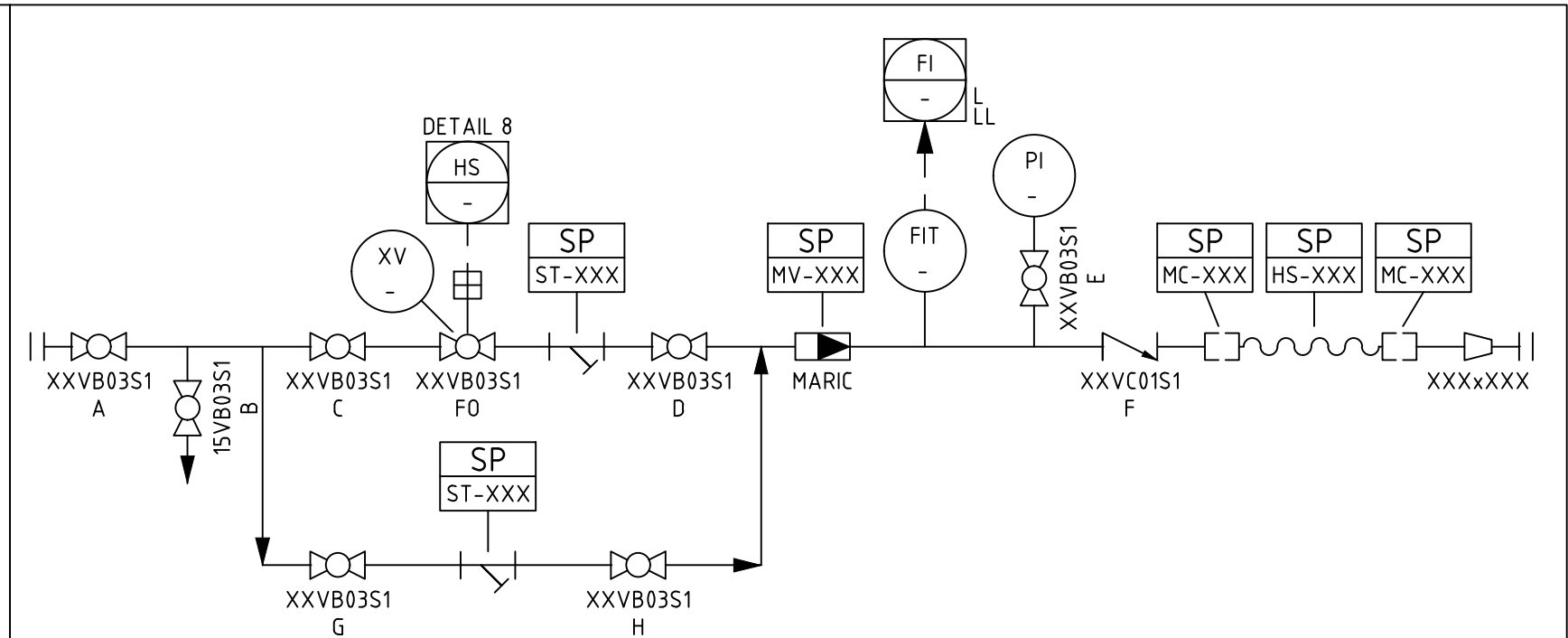
- REFER TO 2000-F24-PID-PR-10003 AND 2000-F24-PID-PR-10014 FOR THE GREENFIELDS LOW PRESSURE GLAND SEAL WATER CIRCUIT.
- REFER TO 2000-F24-PID-PR-10004 AND 2000-F24-PID-PR-10014 FOR THE GREENFIELDS HIGH PRESSURE GLAND SEAL WATER CIRCUIT.
- PUMPS REQUIRING GLAND WATER THAT ARE WITHIN BROWNFIELDS SCOPE ARE TO TIE IN TO THE EXISTING GLAND WATER CIRCUIT. REFER TO 2000-F75-PID-PR-10007 FOR THE BROWNFIELDS GLAND WATER CIRCUIT.
- VALVE SIZE MATCHES LINE SIZE UNLESS OTHERWISE SPECIFIED IN THE RELEVANT GLAND WATER DETAIL.
- LP REFERS TO THE GREENFIELDS LOW PRESSURE GLAND WATER CIRCUIT, HP REFERS TO THE GREENFIELDS HIGH PRESSURE GLAND WATER CIRCUIT AND BF REFERS TO THE BROWNFIELDS GLAND WATER CIRCUIT.
- REDUCING BUSH REQUIRED BETWEEN HOSE AND PUMP CONNETION.



GLAND WATER DETAIL 1
SOLENOID VALVE AND FLOW SWITCH
(REFER TABLE FOR SIZING AND INFORMATION)



GLAND WATER DETAIL 2
SOLENOID VALVE AND FLOW TRANSMITTER (DN25 & UNDER)
(REFER TABLE FOR SIZING AND INFORMATION)



GLAND WATER DETAIL 3
BALL VALVE AND FLOW TRANSMITTER (DN32 & ABOVE)
(REFER TABLE FOR SIZING AND INFORMATION)

EQUIPMENT DETAILS						MANUAL VALVE TAG NUMBERS								INSTRUMENTATION			SPECIAL PIPING ITEMS					
EQUIPMENT NUMBER	PUMP DESCRIPTION	P&ID NUMBER	LINE SIZE NOTE 4	DETAIL	HP/LP/BF NOTE 5	A	B	C	D	E	F	G	H	XV/SV/HS	FSL/FAL	PI	SP-MV	SP-ST	SP-MC	SP-HS	REDUCER	
F21-PP-478	CIL4 TANK 1 OXYGEN INJECTION PUMP	2000-F21-PID-PR-10001	25	1	LP	F21-10570	F21-10571	F21-10572	F21-10573	F21-10575	F21-10576	F21-10574	-	21320	21321	21322	002	011	001	015	N/A	
F21-PP-491	CIL4 TANK 2 OXYGEN INJECTION PUMP	2000-F21-PID-PR-10001	25	1	LP	F21-10577	F21-10578	F21-10579	F21-10580	F21-10582	F21-10583	F21-10581	-	21324	21325	21326	002	011	001	015	N/A	
F22-PP-490	CIL4 BARREN CARBON PUMP	2000-F22-PID-PR-10005	25	1	LP	F22-10584	F22-10585	F22-10586	F22-10587	F22-10589	F22-10590	F22-10588	-	22260	22261	22262	001	011	001	015	25x8	
F28-PP-461	UFG MILL 3 FEED PUMP	2000-F28-PID-PR-10003	25	1	HP	F28-11921	F28-11922	F28-11923	F28-11924	F28-11926	F28-11927	F28-11925	-	28433	28434	28435	005	011	001	015	25x8	
F28-PW-431	UFG MILL 3 ISACHARGER PUMP	2000-F28-PID-PR-10003	25	1	HP	F28-11935	F28-11936	F28-11937	F28-11938	F28-11940	F28-11941	F28-11939	-	28430	28431	28432	002	011	001	015	25x10	
F28-PP-463	UFG MILL 3 DISCHARGE PUMP	2000-F28-PID-PR-10004	25	1	HP	F28-11907	F28-11908	F28-11909	F28-11910	F28-11912	F28-11913	F28-11911	-	28436	28437	28438	004	011	001	015	25x15	
F72-PP-658	SLAKED LIME TRANSFER PUMP 1	2000-F72-PID-PR-10001	25	1	HP	F72-10774	F72-10775	F72-10776	F72-10777	F72-10779	F72-10780	F72-10778	-	72492	72493	72494	005	011	001	015	25x8	
F72-PP-659	SLAKED LIME TRANSFER PUMP 2	2000-F72-PID-PR-10002	25	1	HP	F72-10809	F72-10810	F72-10811	F72-10812	F72-10814	F72-10815	F72-10813	-	72495	72496	72497	005	011	001	015	25x8	
F72-PP-656	LIME RINGMAIN PUMP 1	2000-F72-PID-PR-10003	25	1	HP	F72-10781	F72-10782	F72-10783	F72-10784	F72-10786	F72-10787	F72-10785	-	72470	72471	72472	005	011	001	015	25x8	
F72-PP-657	LIME RINGMAIN PUMP 2	2000-F72-PID-PR-10003	25	1	HP	F72-10788	F72-10789	F72-10790	F72-10791	F72-10793	F72-10794	F72-10792	-	72474	72475	72476	005	011	001	015	25x8	
F30-PP-415	CONCENTRATE PRE-LEACH THICKENER U/F PUMP 1	2000-F30-PID-PR-10004	25	1	BF	F30-10598	F30-10599	F30-10600	F30-10601	F30-10603	F30-10604	F30-10602	-	30403	30404	30405	001	011	001	015	25x8	
F30-PP-416	CONCENTRATE PRE-LEACH THICKENER U/F PUMP 2	2000-F30-PID-PR-10004	25	1	BF	F30-10605	F30-10606	F30-10607	F30-10608	F30-10610	F30-10611	F30-10609	-	30408	30409	30410	001	011	001	015	25x8	
F30-PP-426	CONCENTRATE LEACH TRANSFER PUMP 1	2000-F30-PID-PR-10006	25	1	BF	F30-10612	F30-10613	F30-10614	F30-10615	F30-10617	F30-10618	F30-10616	-	30434	30435	30436	005	011	001	015	25x8	
F30-PP-427	CONCENTRATE LEACH TRANSFER PUMP 2	2000-F30-PID-PR-10006	25	1	BF	F30-10619	F30-10620	F30-10621	F30-10622	F30-10624	F30-10625	F30-10623	-	30440	30441	30442	005	011	001	015	25x8	
F30-TK32A-PP01	CIL2 TANK 1 OXYGEN INJECTION PUMP	2000-F30-PID-PR-10007	25	1	BF	F30-10563	F30-10564	F30-10565	F30-10566	F30-10568	F30-10569	F30-10567	-	30552	30553	30554	002	011	001	015	N/A	
F30-TK32B-PP01	CIL2 TANK 2 OXYGEN INJECTION PUMP	2000-F30-PID-PR-10007	25	1	BF	F30-10795	F30-10796	F30-10797	F30-10798	F30-10800	F30-10801	F30-10799	-	30567	30568	30569	002	011	001	015	N/A	
F30-PP-428	CIL3 TAILS PUMP 1	2000-F30-PID-PR-10011	25	1	BF	F30-10626	F30-10627	F30-10628	F30-10629	F30-10631	F30-10632	F30-10630	-	30466	30467	30468	005	011	001	015	25x8	
F30-PP-429	CIL3 TAILS PUMP 2	2000-F30-PID-PR-10011	25	1	BF	F30-10633	F30-10634	F30-10635	F30-10636	F30-10638	F30-10639	F30-10637	-	30472	30473	30474	005	011	001	015	25x8	
F65-PP-481	CIL3 LOADED CARBON PUMP	2000-F30-PID-PR-10017	25	1	BF	F65-10802	F65-10803	F65-10804	F65-10805	F65-10807	F65-10808	F65-10806	-	65408	65409	65410	001	011	001	015	25x8	
F34-PP-443	UFG MILL 1 PRODUCT PUMP	2000-F34-PID-PR-10002	25	1	BF	F34-10760	F34-10761	F34-10762	F34-10763	F34-10765	F34-10766	F34-10764	-	34406	34407	34408	001	011	001	015	25x8	
F35-PP-449	UFG CIRCUIT 3 FEED PUMP	2000-F35-PID-PR-10002	25	1	BF	F35-10767	F35-10768	F35-10769	F35-10770	F35-10772	F35-10773	F35-10771	-	35445	35446	35447	001	011	001	015	25x8	

2000-F00-PID-PR-10007	LEGEND SHEET 7 - PIPING AND INSTRUMENTATION DIAGRAM																						
2000-F00-PID-PR-10006	LEGEND SHEET 6 - PIPING AND INSTRUMENTATION DIAGRAM																						
2000-F00-PID-PR-10005	LEGEND SHEET 5 - PIPING AND INSTRUMENTATION DIAGRAM																						
2000-F00-PID-PR-10004	LEGEND SHEET 4 - PIPING AND INSTRUMENTATION DIAGRAM																						
2000-F00-PID-PR-10003	LEGEND SHEET 3 - PIPING AND INSTRUMENTATION DIAGRAM																						
2000-F00-PID-PR-10002	LEGEND SHEET 2 - PIPING AND INSTRUMENTATION DIAGRAM																						
2000-F00-PID-PR-10001	LEGEND SHEET 1 - PIPING AND INSTRUMENTATION DIAGRAM																						
DRG No.	REFERENCES	REV	DATE	DESCRIPTION	DRAW	CHECK	DESIGN	TECH APP	PROJ APP														
		1	12SEP25	SPECIAL PIPING ITEMS ADDED	AH	MM	JBE	LV	DM														
		0	27JUN25	APPROVED FOR CONSTRUCTION	AH	JDM	JBE	LV	DM														



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PROJECT:	NORTHERN STAR RESOURCES LTD KCGM GROWTH PROJECT			
TITLE:	AIR & WATER SERVICES GLAND WATER - SHEET 5 PIPING AND INSTRUMENTATION DIAGRAM			
SHEET	A1	SCALE	NTS	REVISION
DRG No.			2000-F24-PID-PR-10035	1